

Task 2. EM algorithm for detective

Course: Bayesian Methods in Machine Learning, Fall 2023

1 Introduction

Lady Margalotta asked the famous detective [Nick Carter](#) for help finding her beloved dog. Even a vast insurance compensation doesn't drown the owner's sorrow. During her trip to London, a stranger stole Lady Margalotta's beloved pet. The CCTV cameras that recorded the thief turned out to be poor assistants, because... exposed to electromagnetic interference. [Group of Bayesian methods](#) is under suspicion.

Nick Carter needs to process the photographs and restore the thief's face. It is known that in various photographs it is located at random coordinates against a stationary background. Fortunately, video lectures on the Bayesian methods course have already been posted online, and after watching them, Nick Carter was able to solve the problem and, using the EM algorithm, identify the thief. He was surprised...

Help the detective find the criminal. Noisy photographs come from the laboratory in portions that will be posted every 2-3 days. An example of a noisy photograph is shown in Figure 1. Don't forget that the faster a criminal is detected, the easier it is to catch him! The person who determined this is asked to promptly inform Nick about this. Information on how to contact him is contained in the «Completing a task» section. The first three students to provide the correct answer will receive thanks from Nick and small prizes.

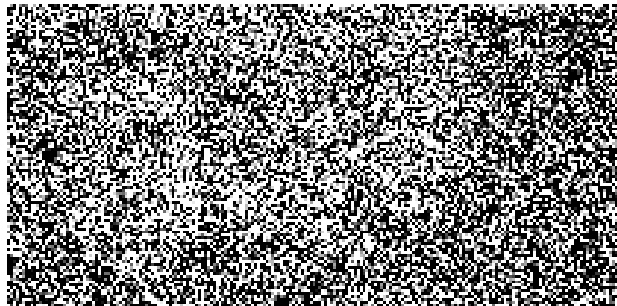


Fig.1: An example of a noisy image from CCTV cameras.

2 Model description

Suppose we are given a sample $\mathbf{X} = \{\mathbf{X}_k\}_{k=1}^K$ of highly noisy black-and-white images of size $H \times W$ pixels. Each of these images contains the same stationary background and the face of the criminal at unknown coordinates, with the entire face included in any image. We will assume that the face image has a rectangular shape of size $h \times w$ pixels. The values of h, w are indicated in the task description in classroom. The layout of the image is shown in Figure 2.

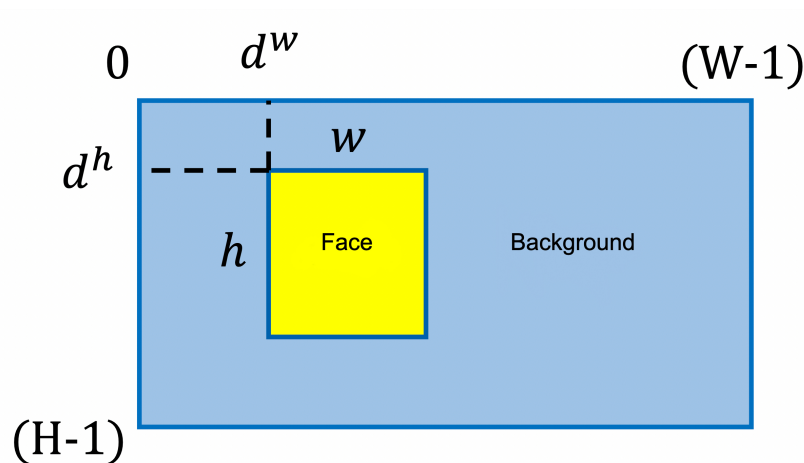


Fig.2: Layout of the image from the camera without noise.

Notation:

- $\mathbf{X}_k(i, j)$ — pixel of the k -th image;
- $\mathbf{B} \in \mathbb{R}^{H \times W}$ — image of a clean background without the face of the criminal, $\mathbf{B}(i, j)$ — pixel of this image;
- $\mathbf{F} \in \mathbb{R}^{h \times w}$ — image of the criminal's face, $\mathbf{F}(i, j)$ — pixel of this image;
- $\mathbf{d}_k = (d_k^h, d_k^w)$ — coordinates of the upper left corner of the face image in the k -th image (d_k^h — vertically, d_k^w — horizontally), $\mathbf{d} = (\mathbf{d}_1, \dots, \mathbf{d}_K)$ — set of coordinates for all images in the sample.

We will also consider the noise in the image to be independent for each pixel and belonging to the normal distribution $\mathcal{N}(0, s^2)$, where s — standard deviation. For one image we have:

$$p(\mathbf{X}_k | \mathbf{d}_k, \boldsymbol{\theta}) = \prod_{ij} \begin{cases} \mathcal{N}(\mathbf{X}_k(i, j) | \mathbf{F}(i - d_k^h, j - d_k^w), s^2), & \text{if } (i, j) \in \text{faceArea}(\mathbf{d}_k) \\ \mathcal{N}(\mathbf{X}_k(i, j) | \mathbf{B}(i, j), s^2), & \text{else} \end{cases},$$

where $\boldsymbol{\theta} = \{\mathbf{B}, \mathbf{F}, s^2\}$, $\text{faceArea}(\mathbf{d}_k) = \{(i, j) | d_k^h \leq i \leq d_k^h + h - 1, d_k^w \leq j \leq d_k^w + w - 1\}$.

Let us set the distribution of the unknown coordinates $\mathbf{d}_k$ to be common for all images using the parameter matrix $\mathbf{A} \in \mathbb{R}^{H-h+1, W-w+1}$:

$$p(\mathbf{d}_k | \mathbf{A}) = \mathbf{A}(d_k^h, d_k^w), \quad \sum_{ij} A(i, j) = 1,$$

where $\mathbf{A}(i, j)$ — matrix element $\mathbf{A}$.

As a result, we have the following joint probabilistic model:

$$p(\mathbf{X}, \mathbf{d} | \boldsymbol{\theta}, \mathbf{A}) = \prod_k p(\mathbf{X}_k | \mathbf{d}_k, \boldsymbol{\theta}) p(\mathbf{d}_k | \mathbf{A}).$$

3 Task formulation

You need to solve a problem

$$p(\mathbf{X} | \boldsymbol{\theta}, \mathbf{A}) \rightarrow \max_{\boldsymbol{\theta}, \mathbf{A}}.$$

Here we can use EM algorithm and optimize the following evidence lower bound:

$$\mathcal{L}(q, \boldsymbol{\theta}, \mathbf{A}) = \mathbb{E}_{q(\mathbf{d})} \log p(\mathbf{X}, \mathbf{d} | \boldsymbol{\theta}, \mathbf{A}) - \mathbb{E}_{q(\mathbf{d})} \log q(\mathbf{d}) \rightarrow \max_{q, \boldsymbol{\theta}, \mathbf{A}}$$

At the E-step, the posterior distribution for coordinates $\mathbf{d}_k$ is calculated for every image:

$$q(\mathbf{d}) = p(\mathbf{d} | \mathbf{X}, \boldsymbol{\theta}, \mathbf{A}) = \prod_k p(\mathbf{d}_k | \mathbf{X}_k, \boldsymbol{\theta}, \mathbf{A}),$$

and at the M-step new parameters $\boldsymbol{\theta}, \mathbf{A}$ are found by solving the following optimization problem:

$$\mathbb{E}_{q(\mathbf{d})} \log p(\mathbf{X}, \mathbf{d} | \boldsymbol{\theta}, \mathbf{A}) \rightarrow \max_{\boldsymbol{\theta}, \mathbf{A}}.$$

We will also consider below a simplified version of the EM algorithm, called hard EM. In this algorithm, after the E-step, it is not necessary to take the entire posterior distribution for the coordinates of the face in the images, but only the MAP estimate for these coordinates (after the E-step $q(\mathbf{d})$ is transformed so that for each image $\mathbf{X}_k$ the estimate $q(\mathbf{d}_k)$ takes the value 1 only at one point — argmaximum point of the posterior distribution $p(\mathbf{d}_k | \mathbf{X}_k, \boldsymbol{\theta}, \mathbf{A})$).

In EM iterations you should monitor the value of the criterion function $\mathcal{L}(q, \boldsymbol{\theta}, \mathbf{A})$.

Theory.

Derive the following formulas:

1. Posterior distribution on face coordinates in images $p(\mathbf{d}_k | \mathbf{X}_k, \boldsymbol{\theta}, \mathbf{A})$ at the E-step;
2. Point estimates for the parameters $\mathbf{A}, \boldsymbol{\theta} = \{\mathbf{F}, \mathbf{B}, s^2\}$ at the M-step for the EM and hard-EM algorithms (note that the point estimates here should be obtained in exactly this order: $\mathbf{A}, \mathbf{F}, \mathbf{B}, s^2$);
3. Evidence lower bound $\mathcal{L}(q, \boldsymbol{\theta}, \mathbf{A})$.

Programming.

Prototypes of all main functions are provided with the assignment. In the assessment the efficiency of the program code will be taken into account. You need to implement:

1. EM algorithm with auxiliary functions with the following stopping criterion:

$$\mathcal{L}(q, \boldsymbol{\theta}^{(t+1)}, \mathbf{A}^{(t+1)}) - \mathcal{L}(q, \boldsymbol{\theta}^{(t)}, \mathbf{A}^{(t)}) < \text{tol},$$

where (t) and $(t+1)$ denote iteration numbers, and tol — a small constant. For the reasons of automatic checking, the following unification of the EM-algorithm procedure is introduced: at each iteration, the E-step must first be launched, then the M-step; outside iterations E and M, steps should not be executed (that is, if the algorithm runs for 1 iteration, steps E and M should be executed exactly 1 time each).

2. Additional functions for performing the M-step and EM algorithm in the case of a hard EM algorithm.
3. A function that runs the EM algorithm several times from different initial approximations.
4. In order to test your implementation, you should generate a sample of small black-and-white noisy images with the same background and some object in a random position. An example of such images before and after noise addition is shown in Figure 3. These images should be small enough to quickly debug the code.

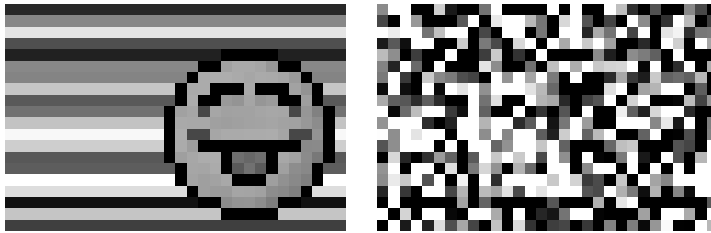


Fig.3: Example images for testing the algorithm before and after noise. Here $h = w = 15, H = 20, W = 30$.

Implementation guidelines:

- The value $\mathcal{L}(q, \boldsymbol{\theta}, \mathbf{A})$ should increase on every iteration. If this is not the case, there is an error in the implementation or in the formulas. This is a good way to debug your program.
- To speed up calculations, $\mathcal{L}(q, \boldsymbol{\theta}, \mathbf{A})$ can be calculated not at every iteration of the algorithm. Use this advice only in your experiments, and send the code for code review with the calculation of this criterion function at each iteration, as written in the prototype.
- In order to avoid problems with the computation accuracy, you should wherever possible switch from products to the sum of logarithms. This point is partially taken into account in function prototypes: the function for calculating $p(\mathbf{X}_k | \mathbf{d}_k, \boldsymbol{\theta})$ must produce logarithmic probability values.
- When calculating $q(\mathbf{d})$, for a more stable normalization it is worth switching to logarithms of probabilities and using the following trick:

$$\alpha_i = \log p_i(\dots) \quad \rightarrow \quad \frac{e^{\alpha_i}}{\sum_k e^{\alpha_k}} = \frac{e^{(\alpha_i - \max_j \alpha_j)}}{\sum_k e^{(\alpha_k - \max_j \alpha_j)}}.$$

- For efficient implementation, you may find it useful to use standard functions that compute convolutions. For example, [fftconvolve](#).

Analysis.

Be sure to provide examples of initial data and results in the report (F, B) .

1. Test the resulting EM algorithm on the generated data. How the initial approximation of the parameters influence the results of the work? Is it worth running the EM algorithm from different initial approximations for this problem?
2. Run the EM algorithm on generated samples of different sizes and with different noise levels. How do changes in the training set affect the results (obtained by F, B and $\mathcal{L}(q, \boldsymbol{\theta}, \mathbf{A})$)? At what noise level does the EM algorithm stop producing normal results? At this point, note that to compare the value of $\mathcal{L}(q, \boldsymbol{\theta}, \mathbf{A})$ for samples of different sizes, it is worth normalizing it to the sample size.

3. Compare the quality and computation time of EM and hard EM using the generated data. Why do you think the difference in work results is so noticeable?
4. Apply the EM algorithm to data with noisy photographs of the criminal. Give the results of the algorithm on samples of different sizes.
5. Suggest some modification of the resulting EM algorithm that would work better and/or faster on this problem.
6. * Implement the proposed modification of the EM algorithm and compare it with the original one in terms of the quality of results and computation time.

The last point (*) is optional. You can get extra points for completing it. Also, additional points will be awarded for the most interesting answers to point 5.

Data Format

Data with noisy photographs of the criminal will be posted every 3 days starting from the day the task was issued (5 times in total). Each time a file will be uploaded that will contain all previously issued data and a new portion of data. The file will have the format `.npy`, that is, it will be a saved `numpy array` of size $H \times W \times K$, where $H \times W$ — is the size of each image, and K — number of images. The values of h, w in the issued data are indicated in assignment's description in classroom.

4 How to submit an assignment

Urgent message to Nick. Message to Nick is optional and does not affect the grade for the assignment. The message should be sent by e-mail to the address nickolas.j.carter@gmail.com with the subject line «CUB - Your Name - Name of the offender».

The letter must indicate who you suspect of the crime. You also need to confirm your suspicions, for example, with a photograph of the criminal that you obtained from your experiments. You can send a message to Nick **only once**. The first three people to send a message with the correct answer will receive thanks from Nick and small prizes.

Assignment completion. For testing in ejudge you need to send a Python module with all the required functions in accordance with the prototypes given in a separate file. The module should be called `name_surname.py`, for example, `petr_ivanov.py`. The module must not contain any `main`! That is, when importing a module, no calculations should be performed.

Before submitting code to ejudge, it must be verified using the public tests provided. If any of them throw a warning (except for timing tests), then your code is not prototype compliant and cannot be verified. Timed warnings indicate that your code is not efficient enough, which may result in a lower score.

To classroom you need to send:

- Python module that was submitted to ejudge. If you implemented the modification as a bonus (point 6 of the analysis), then also attach the modification code.
- A PDF report indicating the last name and first name, containing a description of all the studies conducted (derivation of the necessary formulas, graphs, analysis and conclusions). The report should not contain code listings or similar things! It is advisable to use latex to compile a report. The file should be called `name_surname.pdf`.

Be careful with the file name format!